

▲ M001 Week 5 — English Worksheet

Topic: First Pyramid · **Course:** Pyramid Problems · **Time:** about 45 minutes

This week the agent **walks the edges** of a square and drops blocks as it moves. Stack those squares, shrink each one, and a pyramid grows.

1 · Predict 🧠

Read each set of steps. Before you imagine the agent doing it, write what you think will happen.

```
set f to 6
place on move ON
repeat 4 times:
  move forward by f
  turn left
```

What shape does the agent draw? How many blocks long is each side?

Why is it **repeat 4 times** and **turn left** ? What would **repeat 3 times** draw instead?


```
set f to 6
repeat 3 times:
  [draw one f × f square]
  move up by 1
  change f by -2
```

List the side length of each layer from bottom to top. Looking from the side, what shape do the stacked squares make?


```
place on move ON  
move forward by 4  
place on move OFF  
move forward by 4
```

Where do the blocks end up? Why does turning **place on move** OFF matter when the agent moves up to start the next layer?

2 · Spot the Bug 🐛

Each block of code below was meant to do something, but it is broken. Read what the code is **supposed** to do, then rewrite it so it works. After that, explain why the original was wrong and why your fix works.

Bug A — The agent is supposed to draw a closed **square** outline.

```
set f to 6
place on move ON
repeat 4 times:
  move forward by f
```

Write the fixed code:

Why was it wrong? Why does your fix work?

Bug B — Each new layer must be **2 blocks smaller** than the one below, so the tower narrows into a pyramid.

```
set f to 8
repeat 4 times:
  [draw one f × f square]
  move up by 1
  change f by 2
```

Write the fixed code:

Why was it wrong? Why does your fix work?

Bug C — Each square must sit one block **higher** than the last.

```
set f to 8
repeat 4 times:
  [draw one f × f square]
  change f by -2
```

Write the fixed code:

Why was it wrong? Why does your fix work?

3 · Tell Me What You Built 📷

Now switch to your homework world. Run your `pyra` command to build a pyramid. Then change the **starting value of `f`** (try a smaller number) and build a shorter one. If the agent ends up facing the wrong way or stuck, use your helper commands to reset it — `1` turns it left, `r` turns it right, and `r1` teleports it back to you.

Send a photo or video of your pyramid, then explain what you did. Use these sentence starters — write 4 to 6 sentences total.

First, I ...

The bottom square has sides ... blocks long.

Each new layer shrinks by ... because ...

place on move matters because ...

The hardest part was ...

If I had more time, I would ...

4 · Record Your Walkthrough 🎥

Now take a video on your phone (or a parent's phone) while you walk the camera around your pyramid. Talk through it like you are teaching someone who has never seen it. Try to use these words: **square, side, layer, shrink, place on move, variable f**.

1. Walk around the pyramid and show all 4 sides.
2. Read your `repeat 4 times` loop out loud and say how it draws one square.
3. Say what `change f by -2` does to each layer.
4. Show one bug you hit and how you fixed it.

Write what you will say in your video. Use the space below to plan it before you record — you can read from it while filming.

Submit ✓

Send this worksheet + your walkthrough video to teacher on KakaoTalk.